

# Implicit Representations of 3D Geometry: A Comprehensive Survey, Benchmark, and Future Directions (*Supplementary Material*)

## A POINT CLOUD RECONSTRUCTION BENCHMARK

### A.1 Evaluation Scenarios

We assemble four evaluation scenarios to cover the diversity of practical reconstruction settings.

**Watertight objects.** We source 42 shapes from the Famous dataset [5] (22 shapes) and the Google Scanned Objects dataset [4] (20 shapes). Since the raw meshes are not guaranteed to be manifold or self-intersection-free, we apply alpha wrapping [13] to all shapes to produce closed, manifold, self-intersection-free geometry suitable for ground-truth evaluation.

**Non-watertight objects.** We select 20 shapes that exhibit two common types of topological irregularity. From the ABC dataset [15], we take 10 CAD assemblies whose intersecting parts introduce non-manifold edges and vertices, testing a method's ability to handle non-manifold topology. From the MGN dataset [18], we take 10 clothed human body scans that contain open boundaries at garment hems and sleeves, testing reconstruction fidelity on surfaces without closed enclosures.

For each object, five point cloud configurations are generated by uniform surface sampling: three noise-free sets at 3K, 30K, and 100K points (termed *Sparse*, *Clean*, and *Dense*), and two noisy variants that add isotropic Gaussian noise with standard deviation  $\sigma \in \{0.002\ell, 0.005\ell\}$  to the 30K configuration, where  $\ell$  denotes the bounding-box diagonal.

**Synthetic scenes.** We use 5 indoor scenes from the SceneNet dataset, preprocessed following POCO [2] to ensure watertight ground-truth meshes, for evaluating scene-level reconstruction under controlled, noise-free conditions.

**Real-scanned scenes.** We adopt 5 scenes from the 3D Scene dataset [18], which retain real acquisition artifacts such as sensor noise, to evaluate robustness under practical scanning conditions.

Scene point clouds are sampled at area-based densities of 500 and 1000 points per  $\text{m}^2$ . In total, the benchmark comprises  $42 \times 5 + 20 \times 5 + 5 \times 2 + 5 \times 2 = 330$  evaluation instances.

### A.2 Evaluated Methods

We evaluate 19 representative methods covering analytical and neural implicit fields. For *analytical fields*, we include SPSR [7] and SAP [12] as baselines. The remaining 17 methods are *neural fields*, divided into two paradigms.

**Learning-based prediction:** ConvONet [11], POCO [2], NDF [3], GIFS [20], GeoUDF [14], and SALS [15].

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ACM 0360-0300/2026/1-ART0

<https://doi.org/XXXXXXX.XXXXXXX>

**Per-instance fitting**, grouped by supervision strategy: direct field supervision (IGR [6], SIREN [16], DiGS [1], NSH [17], StEik [19], S<sup>2</sup>DF [18]), projection-based supervision (Neural-Pull [10], I-Filtering [9], LevelSetUDF [22], CAP-UDF [21]), and PG-SDF [8] which combines both.

Among all methods, NDF, GIFS, GeoUDF, SALS, LevelSetUDF, CAP-UDF, and S<sup>2</sup>DF support open-surface reconstruction; the remaining methods inherently produce closed surfaces and are therefore evaluated only on watertight objects and synthetic scenes. Note that I-Filtering also provides a UDF formulation; we adopt its SDF variant in this benchmark and accordingly include it only in the watertight and synthetic-scene evaluations.

SPSR requires oriented normals by design; we additionally evaluate IGR with oriented normals to examine the benefit of orientation information; all other methods operate on unoriented point clouds. SAP is evaluated in its generalization variant (SAP-Gen) on objects and its optimization variant (SAP-Opt) on scenes, as SAP-Gen generalizes poorly to large-scale scenes while SAP-Opt underperforms SAP-Gen on objects.

### A.3 Implementation and Evaluation Protocol

All methods extract meshes on a  $256^3$  grid. For open-surface methods, all methods except SALS use double covering for surface extraction, while other methods follow their default extraction pipelines. Learning-based methods and SAP-Gen are run with the pretrained models invoked by their default inference code; Table 1 lists the corresponding model identifiers used in our benchmark. For per-instance fitting methods, we standardize the optimization budget by supervision type: direct field supervision methods run for 10K gradient steps on objects and 50K on scenes, projection-based methods run for 40K and 80K steps, respectively, and PG-SDF follows its default progressive schedule. ConvONet, Neural-Pull, and GeoUDF are excluded from scene-level evaluation, as their time or memory costs become prohibitive with increasing scene extent on a single GPU.

Table 1. Pretrained models used by learning-based and generalization-based methods.

| Method                 | Pretrained model                                         |
|------------------------|----------------------------------------------------------|
| ConvONet (object) [11] | shapenet_3plane                                          |
| SAP-Gen (object) [12]  | ours_noise_005                                           |
| POCO (object) [2]      | ShapeNet_None_FKACnv_InterpAttentionKHeadsNet_None       |
| POCO (scene) [2]       | SyntheticRooms_None_FKACnv_InterpAttentionKHeadsNet_None |
| NDF [3]                | shapenet_cars_pretrained                                 |
| GIFS [20]              | checkpoint_44h_0m_56s_158456.50156092644                 |
| GeoUDF [14]            | log_reconstruction_100.00_1.000_0.100                    |
| SALS [15]              | logs_thingi10k/01_24_23_02_00                            |

For evaluation, all benchmark instances are preprocessed into a normalized coordinate frame. We uniformly sample  $10^5$  surface points from both the predicted and ground-truth meshes, and normalize both point sets using the center and maximum side length of the ground-truth surface samples, yielding a common unit-scale frame. For real-scanned scenes, predicted faces farther than 0.008 from the input scan are removed before sampling. We report four metrics:

**Chamfer Distance** (CD,  $\downarrow$ , L1,  $\times 10^3$ ) measures the average surface deviation between the prediction and the ground truth. For each predicted point we find its nearest neighbor on the ground-truth surface and record the L1 distance, and vice versa. The reported value is the sum of the two resulting means.

**F-Score** ( $F1$ ,  $\uparrow$ ,  $\%$ ,  $\tau = 0.004$ ) captures the fraction of surface that is accurately reconstructed. A predicted point is counted as a true positive if its nearest ground-truth neighbor is closer than  $\tau$ , yielding precision; a ground-truth point is counted as recovered if its nearest predicted neighbor falls within the same threshold, yielding recall. We report the harmonic mean of the two.

**Normal Consistency** (NC,  $\uparrow$ ,  $\%$ ) evaluates how well surface orientations are preserved. For each nearest-neighbor pair identified in the Chamfer Distance computation, we take the absolute cosine similarity between their normals. The reported value is the average across both matching directions.

**Hausdorff Distance** (HD,  $\downarrow$ , 97th-percentile,  $\times 10^3$ ) quantifies the worst-case surface error. In each matching direction we compute the 97th percentile of the nearest-neighbor distances, and report the larger of the two.

To ensure reproducibility, we implement all benchmarked methods within a unified evaluation pipeline. Each method is wrapped behind the same point-cloud-to-mesh interface, with dependencies isolated in separate micromamba environments. A companion script evaluates all reconstructed meshes in batch and automatically populates the result tables.

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